$$P(A) = \frac{2}{40},\ P(B/A) = \frac{2}{39}$$

$$P(A \cap B) = P(A) \,.\, P(B/A)$$

$$= \frac{2}{40}\,\frac{2}{39}$$

$$= \frac{1}{390}\ .$$

Example 12.19

A coin is tossed twice. Events E and F are defined as follows

E= Head on first toss, F = Head on second toss. Find

(i) $P(E \cup F)$ (ii) $P(E / F)$

(iii) $P(\overline{E} / F)$. (iv) Are the events E and F independent?

Solution

The sample space is $S = \{H, T\} \times \{H, T\}$

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

$$\text{and } E = \{(H, H), (H, T)\}$$

$$F = \{(H, H), (T, H)\}$$

$$E \cup F = \{(H, H), (H, T), (T, H)\}$$

$$E \cap F = \{(H, H)\}$$

(i) $$P(E \cup F) = P(E) + P(F) - P(E \cap F) \quad \text{or} \left(= \frac{n(E \cup F)}{n(S)}\right)$$

$$= \frac{2}{4} + \frac{2}{4} - \frac{1}{4} = \frac{3}{4}\ .$$

(ii) $$P(E / F) = \frac{P(E \cap F)}{P(F)} = \frac{(1/4)}{(2/4)} = \frac{1}{2}$$

(iii) $$P(\overline{E} / F) = \frac{P(\overline{E} \cap F)}{P(F)}$$

$$= \frac{P(F) - P(E \cap F)}{P(F)}$$

$$= \frac{(2/4) - (1/4)}{(2/4)}$$

$$= \frac{1}{2}$$

(iv) Are the events E and F independent?

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